

TASK 1: FORMULATING THE WORLD AS AN MDP

Scenario: Smart Traffic Signal Control

A smart traffic signal system controls traffic flow at a busy intersection. The objective is to reduce traffic congestion while maintaining smooth traffic movement and pedestrian safety.

State Space (S)

State	Meaning
Light	Low traffic
Moderate	Normal traffic flow
Heavy	High traffic congestion

Action Space (A)

Action	Meaning
KeepGreen	Continue current green signal
SwitchSignal	Change traffic direction
PedestrianMode	Enable pedestrian crossing

Reward Function (R)

Situation	Reward
Smooth traffic flow	+10
Reduced congestion	+6
Pedestrian crossing success	+4
Heavy congestion	-8

Manual Value Iteration

Discount Factor: $\gamma = 0.9$

Initial Values: $V_{\blacksquare}(s)=0$

After two iterations, the updated values are calculated.

Final Value Table

State	$V_{\blacksquare}$	$V_{\blacksquare}$
Light	9.2	17.264

Moderate	8	15.74
Heavy	3.2	9.536

Conclusion

The Smart Traffic Signal Control problem was successfully modeled as a Markov Decision Process (MDP). Using value iteration, the optimal state values improved after each iteration, demonstrating how reinforcement learning can optimize real-world traffic systems.